City of Cambridge
Office of Sustainability

Net Zero Action Plan

Fiscal Year 2025 Update
March 31, 2026

CITY OF CAMBRIDGE
Office of Sustainability

Agenda

- Overview of the Net Zero Action Plan (NZAP)
- FY25 update, FY26 progress-to-date and next steps
- State of the World – state and federal actions on climate



Agenda

- **Overview of the Net Zero Action Plan (NZAP)**
- FY25 update, FY26 progress-to-date and next steps
- State of the World – state and federal actions on climate

- How is Cambridge working on climate?
- What is the Net Zero Action Plan?
- How is the Plan reviewed and updated?

How is Cambridge working on climate?

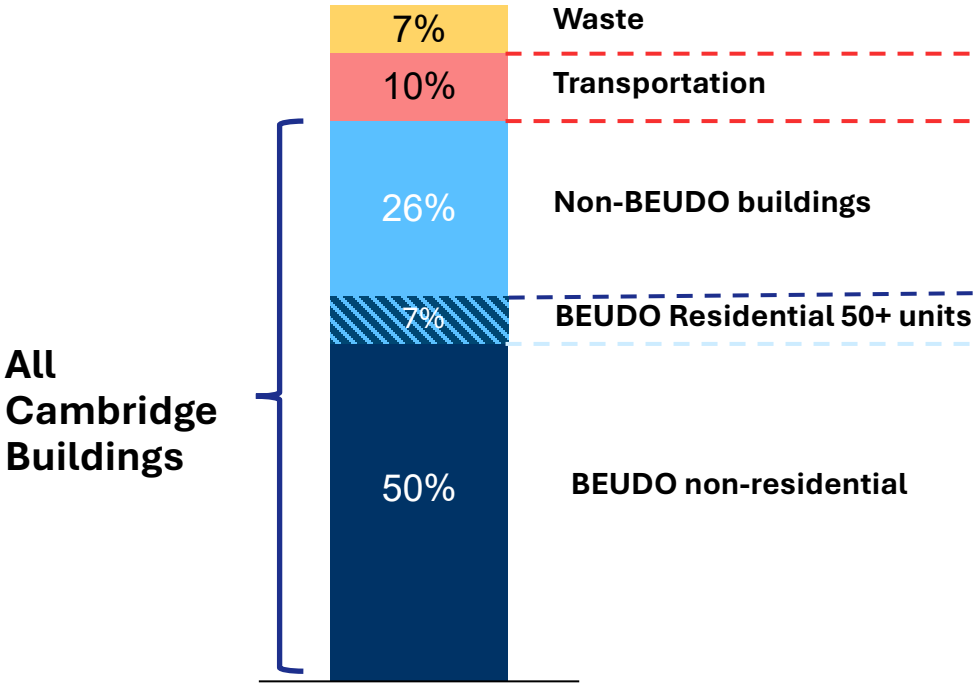
- We work to eliminate greenhouse gas emissions by 2050 or sooner
- We work to prepare Cambridge for the impacts of climate change
- Key frameworks guide this work:
  - Net Zero Action Plan (building emissions)
  - Zero Waste Master Plan
  - Net Zero Transportation Plan
  - Resilient Cambridge Plan
  - Hazard Mitigation Plan
- The Office of Sustainability leads the implementation of the Net Zero Action and oversees Resilient Cambridge



What is the Net Zero Action Plan?

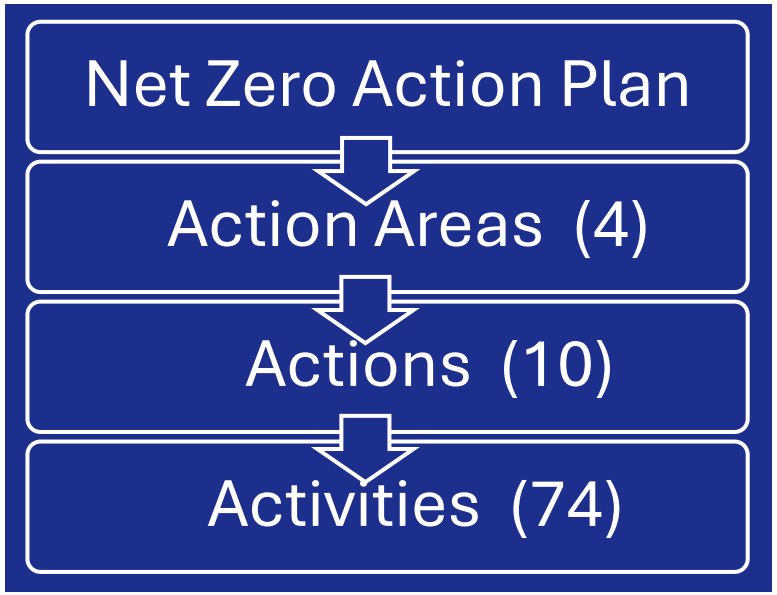
- Cambridge's plan to eliminate greenhouse gas (GHG) emissions from buildings by 2050
- 80% of our greenhouse gas emissions are from buildings
- As a result, the NZAP is the largest component of achieving our climate goals
- First adopted in 2015 and updated every five years

Cambridge's Carbon Emissions (2019)



What does the Plan encompass?

- The Net Zero Action Plan covers four key areas:
  - Energy Efficiency in **Existing Buildings**
  - Net Zero **New Construction**
  - Low Carbon **Energy Supply**
  - **Financing** and Capacity Building
- Each Action Area includes actions, and activities for each action
  - Current timeline is through 2030
- Specific, Measurable, Achievable, Relevant and Time-bound (SMART) goals for each Action added in 2023.



Structure of the Net Zero Action Plan

How is the Plan structured?



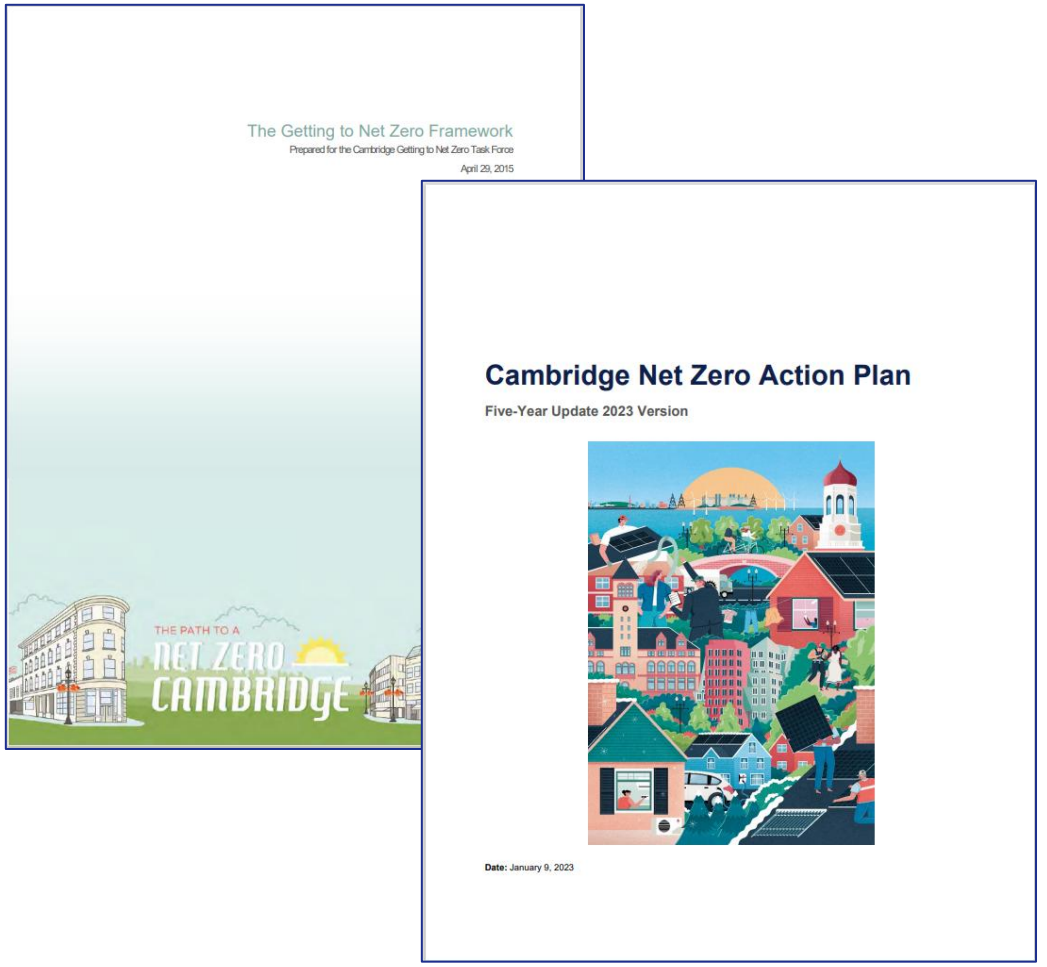
Equity and Resilience in the NZAP

- The current NZAP identifies how actions that reduce GHG emissions also can improve equity and resiliency.
- Examples of actions that can **enhance** equity and resilience include:
  - Programs designed to help support low-to-moderate income residents
  - Building retrofit programs that also include advising on resilience measures.

Action	Equity	Resilience
1.1 Custom Retrofit Program		
1.2 BEUDO Performance Requirements		
1.3 Transaction Points		
2.1 Net Zero Requirements for New Construction		

How is the Plan reviewed and updated?

- **Annual reporting** on the progress of the NZAP by OOS staff
- The Plan requires oversight and annual review by the advisory Climate Committee
- The NZAP is required to be **updated every 5 years** through a full review and community-based process.
- The last update process was conducted in 2021, with Council adoption in 2023.

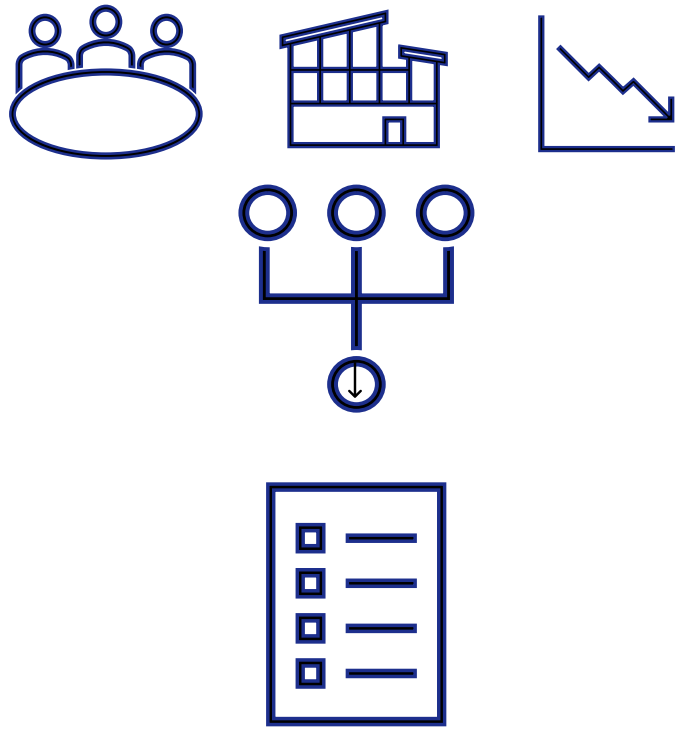


How is the Plan reviewed and updated?

- The required **five-year update** takes into account many changes:
  - Outcomes of NZAP actions implemented to date
  - Evolution of technology (e.g., building electrification)
  - New state and federal policy and other regulatory developments
  - Changing costs and economics
  - Changing stakeholder needs
- The update provides a framework for **checking progress and adjusting** our climate work, similar to the Commonwealth's 5-year climate plan updates
- OOS will begin the next five-year update in **spring 2026**

How is the Plan reviewed and updated?

- The **2026 update** will include:
  - Review changing conditions since last Plan update
  - Projection of emissions outcomes in 2030 and 2050
- The City will conduct an equitable **community engagement** process
  - Gather input on questions of energy affordability, scalability, and equity
- This input and additional research will help develop updated actions, extending beyond 2030



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- State of the World – state and federal actions on climate

- Key Accomplishments for Cambridge
- Review of Action Areas:
  - FY25 Review
  - SMART Goals
  - FY26 progress-to-date
  - Next Steps

Cambridge is implementing leading-edge climate action

New Buildings

- As one of the first, Cambridge adopted the **Specialized Stretch Code** in 2023, requiring all new buildings to be highly energy efficient.
  - 57 Massachusetts municipalities have now adopted the SSC
- Further, Cambridge is one of 10 **Fossil Fuel Free** communities in Massachusetts allowed to require nearly all new buildings, major renovations, and building additions to only use all-electric energy systems.

Existing Buildings

- Starting in January 2026 the **BEUDO amendments** require large, non-residential buildings to begin reducing emissions on a set schedule:
  - BEUDO emissions represent 50% of total GHG emissions in Cambridge

Cambridge is implementing leading-edge climate action

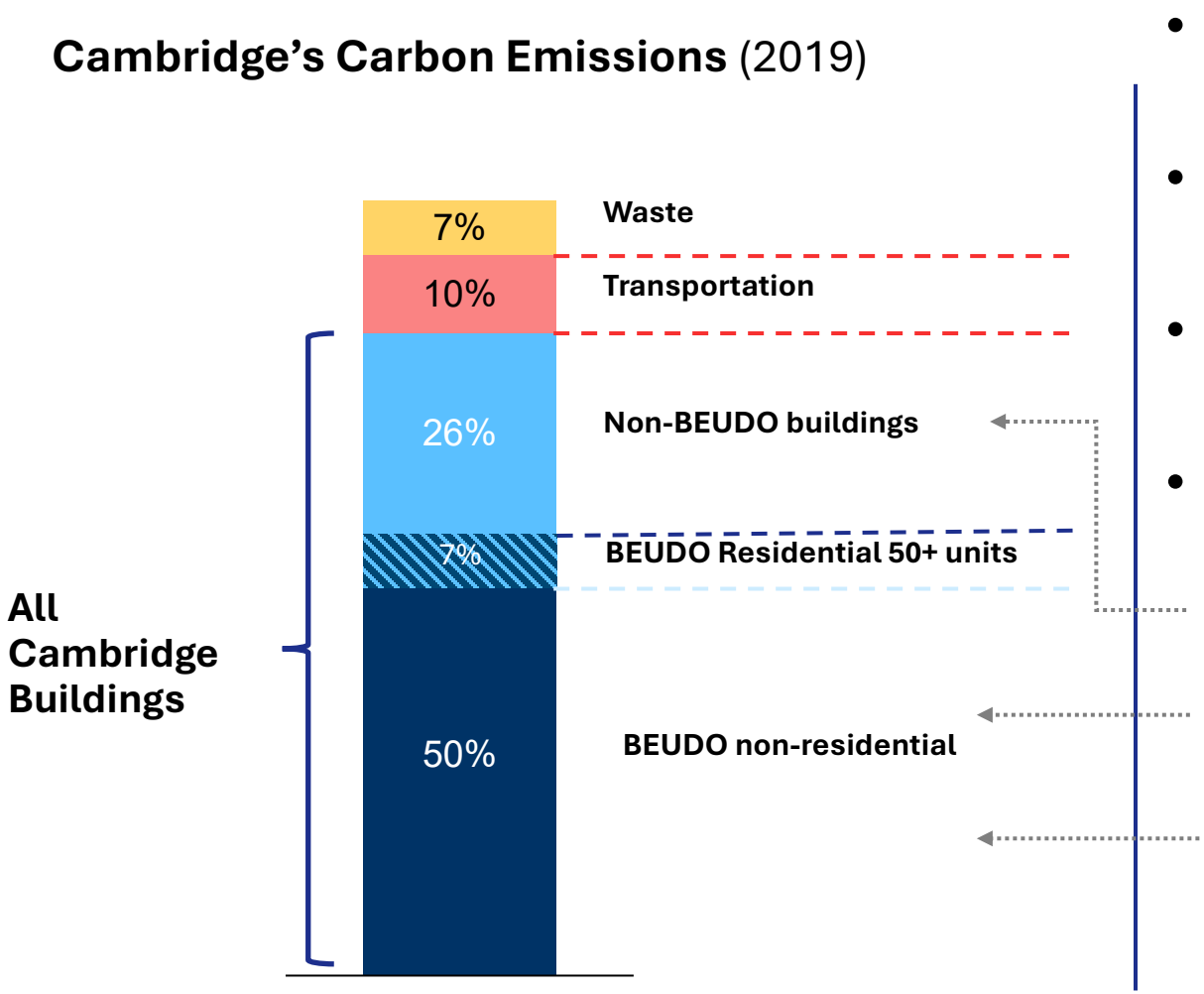
Residential Buildings and Small Businesses

- The **Electrify Cambridge** program has provided no-cost advising on all-electric solutions to over 1,359 households, and now includes **cool roofs** support
- The **Green Equipment Program** (ARPA funded) is on track to help 20 businesses decarbonize their operations through heat pumps, kitchen equipment, etc.
- The number of Cambridge households getting **home energy assessments, weatherization, and heat pumps** is at a historic high. The City programs provide energy bill checkups and connect people to clean energy and efficiency programs.

Clean Energy Supply

- Cambridge Community Electricity serves over 42,000 customers and now offers a default of **75% green electricity** thanks to a historic power purchase agreement.

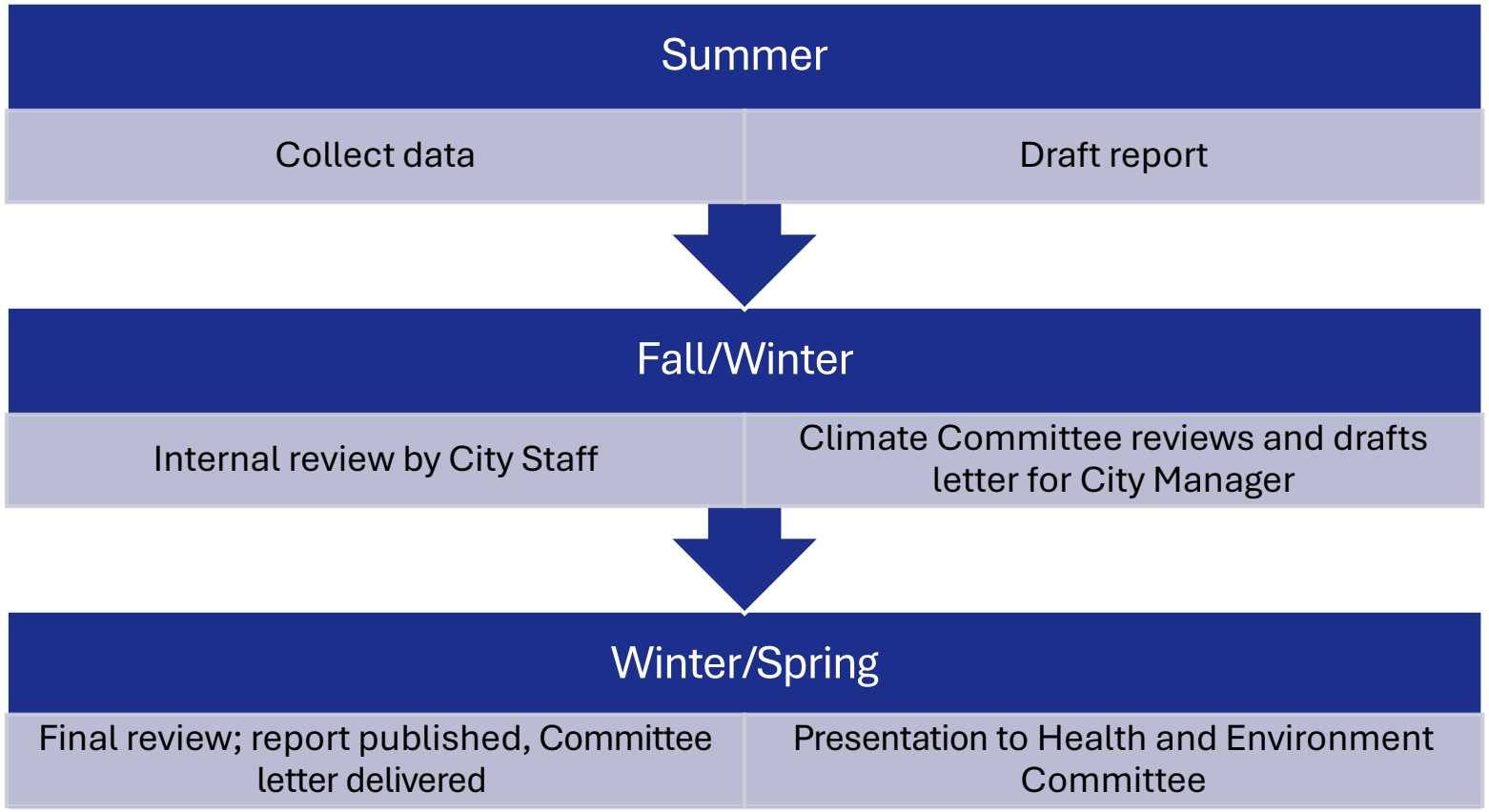
Key emissions reductions expected in the coming years



- Cambridge's GHG Inventory examines the sources and trends of our emissions
- Updating the inventory is the first step in our five-year NZAP review
- Data come from many sources; inventory being completed this spring
- Near term impacts 2026-2030:
  - Renewable electricity from the VPPA lowers residential/small business GHG
  - The largest BEUDO non-res. buildings to achieve 20% lower emissions in 2026-29
  - Other BEUDO non-res. buildings start reducing emissions in 2030
  - Major renovations to be fossil fuel free

Review Process: Fiscal Year Reporting

This is the intended timeline for preparing and reviewing the annual NZAP FY report.



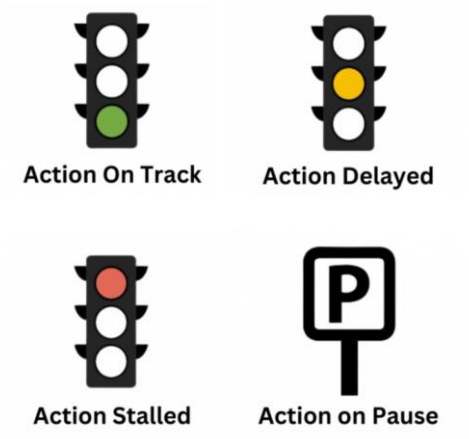
Review Process: Fiscal Year Reporting



- Fiscal Year NZAP Report provides high-level coverage of the previous fiscal year
- Insights and updates reported at the Action-level
- Activity-level status is provided as part of the NZAP report as interactive, drop-down feature on the website



Activity-level Status Indicators

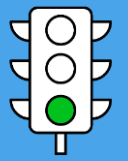
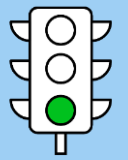
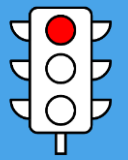


- **The green light** indicates that the action is on track with the timeline adopted in 2023.
- **The yellow light** indicates that the action is delayed from 2023 timeline but making progress.
- **The red light** indicates the action is stalled and is not currently making progress.
- **The parking sign** indicates action is on pause.

For each Action Area, we cover:

1. FY25 Status of each Action
2. Status of each SMART goal
3. Activity in FY26
4. Next Steps

FY25 Status for Action Area 1: Energy Efficiency in Existing Buildings

Action	Description	Status
Action 1.1 Custom Retrofit Program	Small and medium residential and small commercial program for energy efficiency	 On Track
Action 1.2 BEUDO Emissions Reductions Requirements	Greenhouse gas reduction standards for buildings over 25,000 square feet or 50+ residential units	 On Track *Requirement to reduce emissions started in 2026 for non-residential buildings. Residential still required to report on annual energy and water use.
Action 1.3 Upgrades at Transaction Points	Greenhouse gas reduction strategies and measures at the time of sale, lease, renovation, or permit application	 Action On Hold

FY25 Action Area 1 SMART Goals: Energy Efficiency in Existing Buildings

Action is progressing

Action is progressing, but slowly

Action is not progressing

Action	SMART Goal	Status	
1.1	750 units engaged in program per year	287 units served	
1.1	10 buildings retrofit through financing pilot	Pilot program closed January 2025; no financing used	
1.1	Develop retrofit tracking approach to set conversion targets by end of 2023	Challenging to measure immediate conversion from consultation to install	
1.2	BEUDO buildings achieve annual emissions reduction	Emissions reduction requirements start: 2026 for nonres buildings >100k sq/ft 2030 for nonres buildings 25-100k sq/ft	
1.2	Resource hub fully established by end of 2023 and annual usage is monitored	MassCEC program and City guidance in development. DOE grant frozen.	
1.3	Adopted transaction-based requirements by June 2024	Focusing on residential retrofit financing needs	
1.3	Advance tools, education, and resource hub as transaction-based requirements are developed	Planning to analyze financing needs and tools	

Action Area 1: Energy Efficiency in Existing Buildings – FY26 Update

- **Residential Custom Retrofit Program**
  - From July 2025 through January 2026, 219 residential units received support from Electrify Cambridge (uptick from same period in previous year).
  - Engaging larger multifamily and condo buildings in Electrify Cambridge remains challenging. Outreach efforts will be expanded over the coming year.
  - In 2025, home energy assessments, weatherization, and heat pump installs reached all-time highs.
  - Upfront electrification project costs remain expensive. OOS beginning Decarbonization Financing Study this spring.

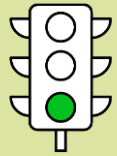
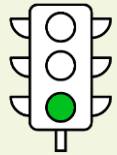
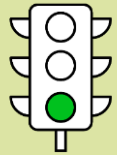
Action Area 1: Energy Efficiency in Existing Buildings – FY26 Update

- **BEUDO Emissions Reduction Requirements**
  - 3 phases of regulations promulgated over the past year. Procedures issued.
  - BEUDO Review Board established and ready to hear cases this spring.
  - Story Map and Technical Guides for BEUDO property owners published.
- Upgrades at **Transaction Points** remained on hold as OOS focuses on decarbonization financing.

Action Area 1: Energy Efficiency in Existing Buildings – Next Steps

- **Residential Custom Retrofit Program**
  - OOS will be expanding direct outreach for Electrify Cambridge and integrating Cool Roof technical assistance.
    - This includes webinars, office hours, and direct engagement with building owners.
    - Technical Advisors follow-up with owners several times offering additional support.
  - Community energy programs will continue to build on the expanded outreach of the past year.
- **BEUDO Emissions Reduction Requirements**
  - Enable access to additional technical and financial resources for BEUDO property owners, particularly for medium-sized buildings subject to 2030 emissions reduction requirements. Maintain high compliance levels.

FY25 Status for Action Area 2: Net Zero New Construction

Action	Description	Status
Action 2.1 Net Zero New Construction	Targets for new construction aligned with state initiatives and standards	 On Track
Action 2.2 Address Embodied Carbon through Green Building Requirements	Address embodied carbon in new construction and major renovations	 On Track
Action 2.3 Net Zero Requirements for Municipal Buildings	Continue commitment to net zero in municipal building projects	 On Track

FY25 Action Area 2 SMART Goals: Net Zero New Construction

Action is progressing

Action is progressing, but slowly

Action is not progressing

Action	SMART Goal	Status	
2.1	Adopt Net Zero Stretch Code by early 2023	Adopted in 2023	
2.1	Adopt Fossil Fuel Free Pilot by summer 2023	Adopted in 2023	
2.1	Prepare resources to support developers for when updates take effect	Implemented in coordination with ISD	
2.2	Adopt carbon reporting requirements for special permit buildings by 2023	Adopted in 2023	
2.3	All new major renovations of municipal buildings are net zero	Major renovations are fossil fuel free	
2.3	Update 2030 GHG reduction goal for municipal buildings by end of 2023	Municipal 2030 GHG target established	

Action Area 2: Net Zero New Construction - FY26 Update

- **Net Zero New Construction**
  - As a result of the Specialized building code and the Fossil Fuel Free Ordinance, new buildings in Cambridge will have significantly lower emissions and, in many cases, no on-site fossil fuel use.
  - Fossil Fuel Free permits issued since March 2024: New Construction – 55, Addition/Alteration – 42

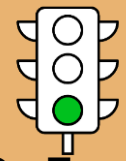
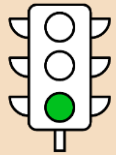
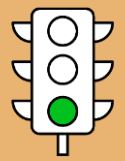
Action Area 2: Net Zero New Construction - FY26 Update

- **Embodied Carbon through Green Building Requirements**
  - Baseline not established as reporting on embodied carbon low; residential development is exempt and few commercial projects have been permitted.
- **Net Zero Requirements for Municipal Buildings**
  - New municipal buildings efficient and fully electric. Through wind and solar farm contracts, 100% of all municipal electricity emissions was eliminated.

Action Area 2: Net Zero New Construction – Next Steps

- **Net Zero New Construction**
  - As part of the Fossil Fuel Free demonstration, Cambridge and the other 9 communities are providing data to the State for potential expansion or development of an opt-in net zero building code.
- **Embodied Carbon**
  - Staff will continue to collect embodied carbon data in order to establish a baseline and set future carbon reduction targets. Cambridge would seek to align with targets implemented elsewhere.
- **Net Zero Requirements for Municipal Buildings**
  - Efficiency and decarbonization projects continue to advance.

FY25 Status for Action Area 3: Low Carbon Energy Supply

Action	Description	Status
Action 3.1 Carbon Free Energy Supply	Continue to support the electrification of buildings. Enable low-carbon district energy.	 On Track
Action 3.2 On-Site Renewable Electricity Access	Promote on-site renewable energy systems and provide support to building owners.	 On Track
Action 3.3 Off-Site Renewable Electricity Access	Provide residents access to renewable energy through the aggregation program. Guide corporate entities into power purchase agreements.	 On Track

FY25 Action Area 3 SMART Goals: Low Carbon Energy Supply

Action is progressing

Action is progressing, but slowly

Action is not progressing

Action	SMART Goal	Status	
3.1	Complete feasibility study for pilot geomicrodistrict by June 2024	Study underway	
3.1	If feasible, install pilot serving at least 500 units plus two public buildings in 2025	Pending feasibility analysis and ownership/financing pathways	
3.1	Establish fossil fuel free conversion targets for buildings by end of 2023	Challenging to measure immediate conversion from consultation to install	
3.2	Achieve 55 MW of solar in Cambridge and on City properties by 2030	Permitted capacity = 13 MW	
3.2	Adopt a solar installation requirement for new construction by June 2023	Green Roof requirement allows solar (Article 22.35.2)	
3.2	Launch a Community Solar program in 2023	Community Solar Administrator Program active	
3.2	Install 5.5 MW of community solar in Cambridge by 2030	Pursing expanded access to LI community solar via CCE	
3.3	Adopted off-site renewable energy criteria through BEUDO in 2023	Regulations published end of 2024	
3.3	Achieve 50% renewable energy for Cambridge Community Electricity Aggregation by 2025, 100% by 2030	Aggregation offers: 100% , 75% (default), and 30% renewable	

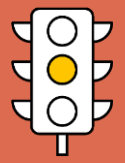
Action Area 3: Low Carbon Energy Supply – FY26 Update

- **Carbon Free Energy Supply**
  - Thermal Energy Network (TEN) Feasibility Study looked at various sites across Cambridge. Focus is on understanding conditions that could make geothermal or other TENs a feasible path to decarbonization.
- **On-Site Renewable Energy**
  - As of the end of 2025, permitted solar capacity in Cambridge stands at 13.1 MW. Program launched 2025 to facilitate large solar projects has had limited impact. Seeking to expand community solar through CCE.
- **Off-Site Renewable Energy**
  - Bowman Wind and Prairie Solar both started generating power in late 2025. CCE default option increased from 50% to 75% green since January 2026. All 3 options now cheaper than Eversource.

Action Area 3: Low Carbon Energy Supply – Next Steps

- **Carbon Free Energy Supply:**
  - Continue to explore Thermal Energy Networks as a strategy. Investigate areas of new development, partnerships with institutions, and regional-scale TEN collaboration.
  - Fundraising support will enable Cambridge Community Center to operate fully off-grid and serve as community resilience center.
- **On-Site Solar:**
  - The Solar Administrator Program will focus on targeting larger roofs as hosts for Community Solar, given changing incentives and economics.
- **Off-Site Solar:**
  - Develop strategy for increasing CCE renewables to 100%.

FY25 Action Area 4: Financing and Capacity Building

Action	Description	Status
Action 4.1 Local Carbon Fund (Enhanced Community Aggregation Program)	Support emissions reductions through offering financial and technical resources.	 Action Delayed

FY25 Action Area 4: Financing and Capacity Building

Action is progressing

Action is progressing, but slowly

Action is not progressing

Action	SMART Goal	Status	
4.1	Adopt Commercial PACE by end of 2023	Adopted Commercial Pace	
4.1	Adopt Local Carbon Fund by 2025	Strategy changed to focus on decarb financing	

Action Area 4: Financing and Capacity Building – FY26 Update

- **Commercial PACE:**
  - The City continues to offer Property Assessment Clean Energy (PACE) funds for commercial and multifamily buildings (adopted in FY25). Statewide program has seen minimal interest.

Action Area 4: Financing and Capacity Building – FY26 Update

- **Local Carbon Fund/Decarbonization Finance Strategy:**
  - Local Carbon Fund determined not to be feasible. Instead, in late 2025 proposals were requested from a qualified consultant team to assist the City in develop decarbonization financing strategies.
  - The strategy will design a long-term financing model to enable residents, small businesses, and nonprofits to implement building-level decarbonization upgrades.
  - Many proposals were received and team to be selected soon.

Action Area 4: Financing and Capacity Building – Next Steps

- **Decarbonization Financing Strategy:**
  - Decarbonization Financing proposals reviewed and contract award expected mid-April. Phases of work include:
    - Examining models for retrofit finance in other cities and regions
    - Determining pathways available to cities in Massachusetts
    - The types of offerings that would be impactful and viable
- **BEUDO Technical Assistance:**
  - Continuing the planning for a technical assistance program for midsize BEUDO buildings to start before 2030, including connecting buildings to financing resources.

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- **State of the World – state and federal actions on climate**
- State of the World

State of the World

- $500 million cut from the MassSave 2025-2027 budget and additional cuts are being contemplated by Legislature
  - Energy affordability framed as a reason to reduce investments in efficiency
  - Cambridge is demonstrating how to cut our emissions, reduce costs, and help our residents
- Development of Massachusetts Decarbonization Clearinghouse and Community Climate Bank
- Statewide Large Building Energy Reporting (LBER)



State of the World

- Energy infrastructure transition: OOS is engaging in state strategy development on grid modernization and gas system transition
- Thermal rate case and legislation would enable utilities to deliver geothermal as a service to large institutions
- Federal impacts to state renewables strategy (e.g., offshore wind) needs to be closely tracked
  - Geothermal tax credit and Direct Pay still available
- Technologies to electrify homes more easily continue to emerge (e.g., new types of heat pumps and induction stoves)



Thank You!